A Divisor-Residue Criterion for Bounded A-Parameters in the Erdős–Straus Conjecture

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Abstract

We present a divisor-residue criterion for the Erdős–Straus conjecture in the residue class $n \equiv 1 \pmod{12}$, $n \not\equiv 0 \pmod 5$. We introduce a parameter $A = 4x - n$ and reduce the conjecture to a **bounded divisor-residue condition**: for each admissible n, there must exist bounded $A \equiv 3 \pmod 4$ and a divisor P of $\big(n(n+A)/4\big)^2$ satisfying

$$P \equiv -n^2 \cdot 4^{-1} \pmod A$$

with positive y, z. We prove that for n prime, $n \equiv 1 \pmod 4$, this condition is equivalent to the target residue T lying in the **bounded divisor-residue set** $D_A((nx)^2)$, a subset of $(\mathbb{Z}/A\mathbb{Z})^*$ generated by prime factors of nx with exponents bounded by $2v_p(nx)$. For A prime, the computational evidence and partial theory connect this condition to a quadratic non-residue criterion via quadratic reciprocity, conditional on a partial -1-route result that is proven in specific subcases and fails in others. We prove unconditionally that prime $n \equiv 5 \pmod 8$ always admits $A = 3$. We also describe a conditional analytic route, via Burgess-type bounds for least prime quadratic non-residues in arithmetic progressions, that would give

$$A = O\Big(n^{1/(4\sqrt{e})+\varepsilon}\Big).$$

Computationally, $A \le 31$ covers all 6,666 cases up to $n = 100{,}000$, and $A \le 99$ covers all but one of 666,666 cases up to $n = 10{,}000{,}000$.

1 Introduction

The Erdős–Straus conjecture states that for every integer $n \ge 2$, there exist positive integers x, y, z such that

$$\frac{4}{n} = \frac{1}{x} + \frac{1}{y} + \frac{1}{z}.$$

Despite extensive work since 1950, the conjecture remains open. Mordell (1967) proved that polynomial identities providing solutions for $n \equiv r \pmod m$ exist only when r is a quadratic non-residue modulo m. Since 1 is a quadratic residue modulo every m, no polynomial identity can cover $n \equiv 1 \pmod m$. This makes $n \equiv 1 \pmod 4$—and in particular $n \equiv 1 \pmod{12}$—the resistant case.

Mballa (2026) introduced a parametric framework claiming density-1 coverage for $n \equiv 1 \pmod 4$ using symmetric solutions, requiring a divisor $b \equiv 3 \pmod 4$ of n. This would handle all composite n having a prime factor congruent to 3 (mod 4), but leaves prime $n \equiv 1 \pmod 4$ as the key open case. Note: Mballa 2026 is a recent preprint; we cite it as a proposed framework pending independent verification.

Our approach does not use polynomial identities. Instead, we parameterize solutions by $A = 4x - n$ and reduce the problem to a **bounded divisor-residue condition**: a specific residue must be achievable as a product of prime powers of nx, with each exponent bounded by the p-adic valuation of nx. This is a concrete combinatorial condition in elementary number theory, not an algebraic-geometric problem.

2 The A-Parameter Framework

2.1 Setup

For $n \equiv 1 \pmod 4$, write

$$x = \frac{n+A}{4}, \qquad A \equiv 3 \pmod 4, \qquad A \geq 3.$$

The Erdős–Straus equation becomes

$$\frac{4}{n} = \frac{4}{n+A} + \frac{1}{y} + \frac{1}{z}.$$

Set

$$m = \frac{n+A}{4}, \qquad nx = n \cdot m.$$

The remainder $4A/(n(n+A))$ must be decomposed as $1/y + 1/z$. The existence of positive y, z is equivalent to finding a divisor P of $(nx)^2$ such that

(i) $P \equiv -n^2 \cdot 4^{-1} \pmod A$, the target residue T; and

(ii) the resulting values

$$y = \frac{P + nx}{A}, \qquad z = \frac{Q + nx}{A}, \qquad Q = \frac{(nx)^2}{P},$$

are positive integers.

We call P a **working divisor** of $(nx)^2$ for the pair (n, A).

2.2 The Bounded Divisor-Residue Set

The set of achievable residues $P \pmod A$, where P ranges over divisors of $(nx)^2$, is

$$D_A((nx)^2) = \left\{ \prod_i p_i^{\alpha_i} \bmod A : 0 \leq \alpha_i \leq 2v_{p_i}(nx) \right\},$$

where the product ranges over all primes p_i dividing nx, and $v_{p_i}(nx)$ is the p_i-adic valuation of nx. This is a **bounded** set: each exponent α_i is limited by $2v_{p_i}(nx)$, reflecting the constraint that P must be an actual divisor of $(nx)^2$.

Note that $D_A((nx)^2)$ is a subset of the subgroup

$$H(A) = \langle p \bmod A : p \mid nx \rangle \subseteq (\mathbb{Z}/A\mathbb{Z})^*,$$

but is generally not equal to $H(A)$, since subgroup membership allows arbitrary integer powers while D_A restricts exponents to bounded ranges. The distinction matters: a subgroup may contain -1 even when no actual divisor of $(nx)^2$ realizes the needed residue.

2.3 The Exact Characterization

Theorem 1 (Divisor-Residue Criterion)**.** *For n prime, $n \equiv 1 \pmod 4$, and $A \equiv 3 \pmod 4$ with* $\gcd(n, A) = 1$*, the parameter A yields an Erdős–Straus solution if and only if*

$$T = -n^2 \cdot 4^{-1} \pmod A$$

lies in $D_A((nx)^2)$.

Proof. By construction, A yields a solution if and only if there exists a divisor $P \mid (nx)^2$ such that

$$P \equiv T \pmod A.$$

Since $m = (n + A)/4 \equiv n \cdot 4^{-1} \pmod A$, we have

$$T = -n^2 \cdot 4^{-1} \equiv -nm \equiv -nx \pmod A.$$

Thus $P \equiv -nx \pmod A$, so $A \mid P + nx$ and hence

$$y = \frac{P + nx}{A}$$

is an integer.

Let $Q = (nx)^2/P$. Because $P \mid (nx)^2$, Q is a positive integer. Also, since $\gcd(nx, A) = 1$, the residue P is invertible modulo A. From $PQ = (nx)^2$ and $P \equiv -nx \pmod A$, we get

$$Q \equiv (nx)^2 P^{-1} \equiv (nx)^2(-nx)^{-1} \equiv -nx \pmod A.$$

Therefore $A \mid Q + nx$, and

$$z = \frac{Q + nx}{A}$$

is an integer. Since $P, Q, nx > 0$, both y and z are positive. Therefore $P \equiv T \pmod A$ with $P \mid (nx)^2$ is exactly the condition for an Erdős–Straus solution arising from this A. □

Remark 1 (Verification). The target may also be written $T = -n \cdot m \pmod A$, since $m \equiv n \cdot 4^{-1} \pmod A$. Computational verification found that $T \in D_A((nx)^2)$ coincides exactly with the existence of Erdős–Straus solutions across all 6,666 test cases with $n \leq 100{,}000$, $n \equiv 1 \pmod{12}$, $n \not\equiv 0 \pmod 5$, with zero mismatches.

Remark 2 (D_A versus $H(A)$). In most tested cases, $D_A((nx)^2) = H(A)$. However, $D_A \neq H(A)$ in general; for example, $n = 7561$, $A = 19$ gives $|D_A| = 6$ but $|H(A)| = 18$. The equivalence $-1 \in D_A \iff -1 \in H(A)$ holds with zero mismatches across all 10,096 prime-A cases where $-1 \in D_A$. However, $-1 \in H(A)$ does NOT always imply $-1 \in D_A$ — 821 counterexamples exist. See the Partial -1 Route below.

3 Main Results

3.1 The $n \equiv 5 \pmod 8$ Case

Theorem 2. *For n prime, $n \equiv 5 \pmod 8$, and $n \neq 3$, the choice $A = 3$ yields an Erdős–Straus solution.*

Proof. When $n \equiv 5 \pmod 8$, we have $n + 3 \equiv 0 \pmod 8$, so

$$m_3 = \frac{n+3}{4}$$

is even. Therefore $2 \mid m_3$, and 2 is a prime factor of $nx = n \cdot m_3$. Since $2 \equiv -1 \pmod 3$, take $P = 2$.

Because n is prime and $n \neq 3$, we have $n^2 \equiv 1 \pmod 3$, and since $4^{-1} \equiv 1 \pmod 3$,

$$-n^2 \cdot 4^{-1} \equiv -1 \equiv 2 \pmod 3.$$

Thus $P = 2$ satisfies the target congruence for $A = 3$. Since $P \mid (nx)^2$, Theorem 1 gives positive integral y, z. Therefore $A = 3$ works directly, without invoking the conditional prime-A criterion. $\square$

3.2 Reduction to a Quadratic Non-Residue Condition

Theorem 3 (Prime A Criterion, conditional on forward direction)**.** *For A prime,* $\gcd(n, A) = 1$*, and n prime with $n \equiv 1 \pmod 4$,*

$$T \in D_A((nx)^2) \quad \Longleftrightarrow \quad \textit{some prime factor } p \mid nx \textit{ is a quadratic non-residue modulo } A.$$

The converse direction is unconditional; the forward direction is conditional and computationally verified in the tested range.

Proof. For A prime, $(\mathbb{Z}/A\mathbb{Z})^*$ is cyclic of order $A - 1$. Since $A \equiv 3 \pmod 4$, $(A - 1)/2$ is odd, and -1 is the unique element of order 2.

For the converse, suppose no prime factor of nx is a quadratic non-residue modulo A. Then all generators of $H(A)$ have odd order, so $h = |H(A)|$ is odd and $-1 \notin H(A)$. Since $D_A \subseteq H(A)$, we have $-1 \notin D_A$. If $T \in D_A$, then using $T = -nm$ and $nm \in H(A)$, we would have

$$T \cdot (nm)^{-1} = -1 \in H(A),$$

a contradiction. Hence $T \notin D_A$.

For the forward direction, if some prime $p \mid nx$ is a quadratic non-residue modulo A, then p has even order, so $h = |H(A)|$ is even and $-1 \in H(A)$. The Partial -1 Route below gives $-1 \in D_A$ in the proven sub-cases ($h = 2$ and order-2 QNR). However, $-1 \in H(A)$ does not always imply $-1 \in D_A$ — 821 counterexamples exist in the tested range. The forward direction is therefore computationally supported but not fully proven.

However, D_A is not closed under multiplication. The step from $-1 \in D_A$ and $nm \in D_A$ to $T = (-1)(nm) \in D_A$ does not follow automatically, because multiplying two divisor residues may push exponents beyond the bound $2v_p(nx)$. If the divisor realizing -1 uses exponent a_i on prime p_i, then the product with nm requires

$$a_i + v_{p_i}(nx) \leq 2v_{p_i}(nx),$$

equivalently $a_i \leq v_{p_i}(nx)$.

Define the shifted divisor-residue set

$$D_A^{(nm)}((nx)^2) = \left\{ \prod_i p_i^{\alpha_i} \bmod A : 0 \leq \alpha_i \leq 2v_{p_i}(nx) - v_{p_i}(nm) \right\}.$$

This is the set of residues R such that $R \cdot nm$ is an actual divisor of $(nx)^2$. The correct sufficient condition for the -1 factorization route is therefore $-1 \in D_A^{(nm)}$, not merely $-1 \in D_A$.

Computationally, $T \in D_A((nx)^2)$ has zero mismatches across all 10,096 prime-A cases tested. In 50 of these cases, $-1 \notin D_A^{(nm)}$, yet $T \in D_A$ via a different divisor path. Thus a complete proof of the forward direction requires a direct argument for $T \in D_A$, or a full proof that $-1 \in D_A^{(nm)}$ when appropriate. □

3.3 The Partial -1 Route

Lemma 1 (Partial -1 Route Result)**.** *For prime A and $\gcd(n, A) = 1$, the implication $-1 \in H(A) \Rightarrow -1 \in D_A((nx)^2)$ holds in the following proven subcases:*

1. *$h = |H(A)| = 2$;*

2. *$H(A)$ contains an order-2 generator $p \equiv -1 \pmod{A}$ among the prime factors of nx.*

Outside these subcases, the implication is not valid in general. In the tested range, 821 cases have $-1 \in H(A)$ but $-1 \notin D_A((nx)^2)$. Nevertheless, the target T may still lie in D_A through alternative divisor paths.

Lemma 2 (Bounded Divisor-Residue, partially proven and computationally verified)**.** *For A prime and $\gcd(n, A) = 1$, if $-1 \in H(A)$, then $-1 \in D_A((nx)^2)$ in the proven sub-cases below and in all tested cases.*

Proof sketch and tested classification. Let $h = |H(A)|$. Since $-1 \in H(A)$, h is even. Let g be a primitive root modulo A, and let $u = g^{(A-1)/h}$ be a generator of $H(A)$. Since $-1 \in H(A)$ and h is even, $-1 = u^{h/2}$ (the unique element of order 2). For each prime factor $p_i \mid nx$, write $p_i \equiv u^{s_i} \pmod{A}$ where $s_i = \mathrm{dlog}_u(p_i)$.

We must show that $h/2$ lies in the bounded sumset

$$S = \left\{ \sum_i s_i \alpha_i \pmod{h} : 0 \le \alpha_i \le 2e_i \right\},$$

where s_i is the discrete logarithm of p_i in $H(A)$ and $e_i = v_{p_i}(nx)$.

If $h = 2$, the nontrivial element of $H(A)$ is $-1 = u^{h/2} = u$. Any quadratic non-residue generator gives $p^1 \equiv -1 \pmod{A}$ with exponent $1 \le 2e$; this case is proven generally. If some prime factor $p \mid nx$ has $\mathrm{ord}_A(p) = 2$, then $p \equiv -1 \pmod{A}$ and again p^1 realizes -1 directly; this case is also proven generally.

For $h = 2m$ with $m > 1$ odd and no order-2 quadratic non-residue, decompose $\mathbb{Z}/h\mathbb{Z} \cong \mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/m\mathbb{Z}$. The $\mathbb{Z}/2\mathbb{Z}$ component is supplied by a single quadratic non-residue. The remaining condition is a bounded sumset condition in $\mathbb{Z}/m\mathbb{Z}$.

Let

$$S' = A_1 + \cdots + A_k \subseteq \mathbb{Z}/m\mathbb{Z},$$

where

$$A_i = \{0, s_i, 2s_i, \ldots, 2e_i s_i\}.$$

If $\mathrm{Stab}(S')$ is trivial and

$$\sum_i |A_i| = \sum_i (2e_i + 1) \ge m + k - 1,$$

then Kneser's theorem gives $|S'| \geq m$, hence $S' = \mathbb{Z}/m\mathbb{Z}$ and the required residue is present. Under the corrected true discrete-log computation, no tested cases currently fall into the trivial-stabilizer Kneser bucket. The condition remains a valid sufficient condition, but it does not currently account for any of the tested cases.

The remaining tested cases are verified computationally. A general proof of the full lemma remains open. □

Case	Count	Status
$h = 2$	1,768	proven generally
order-2 QNR ($h > 2$)	1,919	proven generally
Kneser, trivial additive stabilizer in $\mathbb{Z}/m\mathbb{Z}$	0	under review (dlog corrected)
Size-only, non-trivial stabilizer	4,608	candidate (not proven)
Computational-only	1,801	computational
Total ($-1 \in D_A$)	10,096	zero failures in tested range
Failures ($-1 \notin D_A$)	821	lemma fails; T may still reach D_A
All prime-A cases	10,917	

Corollary 1 (Conditional sufficient condition)**.** *If $\left(\frac{A}{n}\right) = -1$, then A works, conditional on the forward direction of Theorem 3. This condition is computationally verified in all tested cases.*

Proof. By quadratic reciprocity, since $n \equiv 1 \pmod 4$,

$$\left(\frac{A}{n}\right) = \left(\frac{n}{A}\right).$$

If $\left(\frac{A}{n}\right) = -1$, then n is a quadratic non-residue modulo A. Since n is a prime factor of nx, the conditional forward direction of Theorem 3 applies. □

3.4 The Composite n Case

Theorem 4 (Mballa 2026, claimed)**.** *If n has a prime factor $p \equiv 3 \pmod 4$, then $A = p$ yields an Erdős–Straus solution.*

This handles all composite $n \equiv 1 \pmod 4$ having a prime factor congruent to $3 \pmod 4$. Combined with Theorem 2, the only remaining case is prime $n \equiv 1 \pmod 8$.

3.5 Legendre Symbol Identity for the m-Route

Theorem 5 (Legendre Symbol Identity)**.** *Let n be prime with $n \equiv 1 \pmod 4$, let A be an odd prime with $A \equiv 3 \pmod 4$, and let p be an odd prime divisor of $m = (n + A)/4$ with $p \nmid A$. Then*

$$\left(\frac{p}{A}\right) = \left(\frac{n}{p}\right).$$

Thus p is a quadratic non-residue modulo A if and only if n is a quadratic non-residue modulo p.

Remark 3. Theorem 5 is stated for prime A. For composite A in the covering set (e.g., $A = 15, 27$), the Legendre symbol should be replaced by the Jacobi symbol; the identity holds computationally but is not stated as a theorem here.

Proof. Since $p \mid m = (n+A)/4$, we have $A \equiv -n \pmod{p}$. By quadratic reciprocity, since $n \equiv 1 \pmod{4}$,

$$\left(\frac{p}{A}\right) = \left(\frac{A}{p}\right)(-1)^{((p-1)/2)((A-1)/2)}.$$

Since $A \equiv 3 \pmod{4}$, $(A-1)/2$ is odd. If $p \equiv 1 \pmod{4}$, then

$$\left(\frac{p}{A}\right) = \left(\frac{A}{p}\right) = \left(\frac{-n}{p}\right) = \left(\frac{-1}{p}\right)\left(\frac{n}{p}\right) = \left(\frac{n}{p}\right).$$

If $p \equiv 3 \pmod{4}$, then

$$\left(\frac{p}{A}\right) = -\left(\frac{A}{p}\right) = -\left(\frac{-n}{p}\right) = -\left(\frac{-1}{p}\right)\left(\frac{n}{p}\right) = \left(\frac{n}{p}\right).$$

Thus the identity holds in both cases. □

Corollary 2 (Route formulation)**.**

Corollary 3 (Conditional)**.** *Assuming the forward direction of Theorem 3, A works if either $\left(\frac{A}{n}\right) = -1$ (the direct route) or some prime $p \mid (n+A)/4$ has $\left(\frac{n}{p}\right) = -1$ (the m-route). Both routes are governed by the same quadratic character of n.*

3.6 Computational Verification

Theorem 6 (Computational, $n \le 100{,}000$)**.** *For all 6,666 integers $n \equiv 1 \pmod{12}$, $n \not\equiv 0 \pmod{5}$, with $13 \le n \le 100{,}000$, there exists*

$$A \in \{3, 7, 11, 15, 19, 23, 27, 31\}$$

yielding an Erdős–Straus solution. The maximum A required is 31.

Theorem 7 (Computational, $n \le 10{,}000{,}000$)**.** *For all 666,666 integers $n \equiv 1 \pmod{12}$, $n \not\equiv 0 \pmod{5}$, with $13 \le n \le 10{,}000{,}000$, there exists $A \le 99$ yielding an Erdős–Straus solution, with one exception: $n = 8{,}803{,}369$, which requires $A = 107$. The maximum A required excluding this outlier is 59.*

The 666,666 count refers to all integers in the residue class, not just primes. Only prime n need be tested after reductions, but the verification included all integers for completeness.

A	Count	Percentage	Cumulative
3	5,533	83.0%	83.0%
7	967	14.5%	97.5%
11	109	1.6%	99.1%
15	20	0.3%	99.4%
19	16	0.2%	99.7%
23	15	0.2%	99.9%
27	1	0.0%	99.9%
31	5	0.1%	100.0%

3.7 Analytic Bound

Proposition 1 (Conditional analytic route via Burgess-type bounds)**.** *Conditional on a suitable least-prime-QNR-in-AP estimate: for n prime, $n \equiv 1 \pmod 4$, $n \geq n_0$, one would obtain a prime $A \equiv 3 \pmod 4$ with $\left(\frac{A}{n}\right) = -1$ and*

$$A \ll n^{1/(4\sqrt{e})+\varepsilon}$$

for any $\varepsilon > 0$. Such A would yield an Erdős–Straus solution by the Corollary to Theorem 3.

Proof. We combine three classical results.

Step 1: Burgess character sum bound. Let χ be a non-principal Dirichlet character modulo a prime q. The Burgess bound [?]; [10, §12.6] states: for any integers $r \geq 3$ and $N \geq 1$,

$$\left| \sum_{k=1}^{N} \chi(k) \right| \leq C_r \cdot q^{1/(4r)} \cdot N^{1-1/r}$$

where C_r depends only on r. For the quadratic character $\chi(\cdot) = \left(\frac{\cdot}{n}\right)$ modulo n, this gives a nontrivial bound when $N > n^{1/(4r)+\delta}$ for any $\delta > 0$.

Step 2: Least quadratic nonresidue. Setting $r = \lfloor 1/(4\varepsilon) \rfloor + 3$ and $N = n^{1/(4\sqrt{e})+\varepsilon}$, the Burgess bound implies

$$\left| \sum_{k=1}^{N} \left(\frac{k}{n}\right) \right| \leq C \cdot n^{1/(4r)} \cdot N^{1-1/r} < N/2$$

for sufficiently large n. Since $\sum_{k=1}^{N} \left(\frac{k}{n}\right) = \#\{\text{QR} \leq N\} - \#\{\text{QNR} \leq N\}$ and the total is N, a bound of $< N/2$ means both QR and QNR sets are nonempty below N. Hence the least quadratic nonresidue modulo n satisfies $n_2(n) \ll n^{1/(4\sqrt{e})+\varepsilon}$ [10, Cor. 12.20].

Step 3: Least prime QNR in arithmetic progression. We need not just any QNR, but a *prime* QNR with $A \equiv 3 \pmod 4$. Consider the character sum restricted to primes:

$$S(X) = \sum_{\substack{p \leq X \\ p \equiv 3 \pmod 4}} \left(\frac{p}{n}\right).$$

By expanding the indicator function of $p \equiv 3 \pmod 4$ via the non-trivial Dirichlet character χ_4 modulo 4 (with $\chi_4(1) = 1$, $\chi_4(3) = -1$), we get

$$S(X) = -\tfrac{1}{2} \sum_{p \leq X} \left(\frac{p}{n}\right) + \tfrac{1}{2} \sum_{p \leq X} \chi_4(p) \left(\frac{p}{n}\right).$$

Each sum is a prime character sum $\sum_{p \leq X} \psi(p)$ where ψ is a non-principal character (either $\left(\frac{\cdot}{n}\right)$ or the product $\chi_4 \cdot \left(\frac{\cdot}{n}\right)$, both non-principal since $n \equiv 1 \pmod 4$). By partial summation, $\sum_{p \leq X} \psi(p)$ relates to $\sum_{k \leq X} \psi(k)$ via:

$$\sum_{p \leq X} \psi(p) = \sum_{k \leq X} \frac{\psi(k)}{\log k} + O(X^{1/2}).$$

Applying the Burgess bound to $\sum_{k \leq X} \psi(k)$ with $X = n^{1/(4\sqrt{e})+\varepsilon}$, we get $S(X) = o(\pi(X; 4, 3))$ for large X. By the prime number theorem in arithmetic progressions [10, Thm 5.7], $\pi(X; 4, 3) \sim X/(2 \log X) \to \infty$. Since $S(X) = o(\pi(X; 4, 3))$, not all primes $p \equiv 3 \pmod 4$ with $p \leq X$ can satisfy $\left(\frac{p}{n}\right) = +1$; there must exist at least one with $\left(\frac{p}{n}\right) = -1$.

Therefore, for $X = n^{1/(4\sqrt{e})+\varepsilon}$ and n sufficiently large, there exists a prime $A \equiv 3 \pmod 4$ with $\left(\frac{A}{n}\right) = -1$ and $A \leq X$. $\square$

Remark 4. *(i)* The exponent $1/(4\sqrt{e})$ arises from optimizing the Burgess bound: for a character modulo q, the nontrivial range begins at $N \approx q^{1/(4\sqrt{e})}$, where the optimal choice $r \to \infty$ yields the exponent $1/(4\sqrt{e}) \approx 0.1515$ [10, §12.6]; [11, §11.3].

(ii) The extension from "least QNR" to "least prime QNR in AP" requires passing from character sums over integers to character sums over primes via partial summation and the PNT. The key point is that the non-principal characters $\left(\frac{\cdot}{n}\right)$ and $\chi_4 \cdot \left(\frac{\cdot}{n}\right)$ are both non-principal (the latter because $n \equiv 1 \pmod 4$ ensures χ_4 and $\left(\frac{\cdot}{n}\right)$ do not cancel), so the Burgess bound applies to both.

(iii) Under GRH, the least quadratic nonresidue modulo n is $O((\log n)^2)$ [12], and the least prime $\equiv 3 \pmod 4$ that is QNR modulo n is also $O((\log n)^2)$.

(iv) Linnik's theorem [13]; [14] guarantees the least prime in any AP $p \equiv a \pmod q$ is $\ll q^L$ for an absolute constant $L \leq 5$, but this is a bound in q (the modulus of the AP), not in n (the modulus of the character). Proposition 8 uses the Burgess bound in n, which is sharper for our application since n is the large parameter.

(v) This proof remains **conditional** in the following sense: while each individual step is rigorously established in the cited literature, the specific combination—Burgess bound applied to the product character $\chi_4 \cdot \left(\frac{\cdot}{n}\right)$ to extract prime QNRs in the progression $p \equiv 3 \pmod 4$—requires careful verification that the error terms combine correctly. We state this as a proposition pending a complete verification of the analytic details by a specialist in analytic number theory.

4 The Covering Cascade

The covering set
$$\{3, 7, 11, 15, 19, 23, 31\}$$
verified for $n \leq 100{,}000$ operates via two mechanisms.

Direct route. For most prime cases up to 100,000, at least one $A \in \{7, 11, 19, 23, 31\}$ has $\left(\frac{A}{n}\right) = -1$, and that A works directly in the tested range. The case $A = 3$ always has $\left(\frac{3}{n}\right) = 1$ when $n \equiv 1 \pmod{12}$.

m-route. For the 21 tested cases where n is a quadratic residue modulo all of $7, 11, 19, 23, 31$, the m-route provides a solution: the smallest quadratic non-residue p of n appears as a prime factor of m_A for some A in the covering set, and $\left(\frac{p}{A}\right) = \left(\frac{n}{p}\right) = -1$ by Theorem 5.

5 The Constant Bound Conjecture

Conjecture 1. *There exists an absolute constant C such that for every prime $n \equiv 1 \pmod 4$, there exists $A \leq C$ with $A \equiv 3 \pmod 4$ and*
$$T \in D_A((nx)^2).$$

Evidence includes:

- $A \leq 31$ for all $n \leq 100{,}000$ in the tested residue class;
- $A \leq 99$ for all but one of 666,666 cases up to $n = 10{,}000{,}000$;
- $A \leq 107$ for all tested cases up to $n = 10{,}000{,}000$;
- a Burgess-type least-prime-QNR-in-AP estimate would give $A = O(n^{0.152\ldots})$, conditionally;
- GRH gives a conditional $O((\log n)^2)$ route.

This conjecture is stronger than what GRH implies. Proving it would require showing that among a finite set of primes, at least one is always a quadratic non-residue modulo n, or proving an equivalent bounded divisor-residue statement.

6 Context from Known Results

6.1 Mordell (1967)

Mordell's theorem rules out polynomial identities for $n \equiv 1 \pmod{m}$. The present framework uses divisor-residue conditions, not polynomial identities.

6.2 Elsholtz–Tao (2013)

Elsholtz and Tao showed that the total number of solutions satisfies an asymptotic of the form $\sum_{p \le N} f(p) \asymp N \log^2 N$. This provides context on solution density.

6.3 Vaughan (1970) and Webb (1970)

Vaughan showed that potential counterexamples have density zero. Webb showed that only a sublinear number of values need more than two terms. These results are consistent with the computational evidence here.

6.4 Bright–Loughran (2020)

Bright and Loughran showed that the Brauer–Manin obstruction does not explain failure of the integral Hasse principle for Erdős–Straus surfaces.

6.5 Mballa (2026)

Mballa introduced a parametric framework claiming density-1 coverage for $n \equiv 1 \pmod{4}$ via symmetric solutions. The present framework gives an exact divisor-residue criterion and a separate unconditional proof for the $n \equiv 5 \pmod{8}$ case. Note: Mballa 2026 is a recent preprint cited as a proposed framework pending independent verification.

6.6 Burgess (1963) and Related Analytic References

The Burgess bound on character sums [**?**]; [10, §12.6] states that for a non-principal Dirichlet character χ modulo q, $|\sum_{k=1}^{N} \chi(k)| \le C_r \cdot q^{1/(4r)} \cdot N^{1-1/r}$ for $r \ge 3$. The classical application gives the least quadratic nonresidue modulo n as $n_2(n) \ll n^{1/(4\sqrt{e})+\varepsilon}$. Proposition 8 extends this to the least *prime* QNR in the progression $p \equiv 3 \pmod{4}$ via partial summation and the PNT in AP.

Additional analytic references:

- **Iwaniec–Kowalski (2004):** *Analytic Number Theory*, AMS Colloquium Publications vol. 53. Standard reference for the Burgess bound (§12.6), PNT in AP (Thm 5.7), and character sum techniques.

- **Montgomery–Vaughan (2007):** *Multiplicative Number Theory I: Classical Theory*, Cambridge University Press. Burgess bound (§11.3) and PNT in AP.

- **Lamzouri–Li–Soundararajan (2015):** Conditional bounds for the least quadratic non-residue under GRH: $n_2(n) \ll (\log n)^2$ [12]. Also bounds for the least prime QNR in AP.

- **Linnik (1944) / Heath-Brown (1992):** Linnik's theorem [13]; [14] guarantees the least prime $p \equiv a \pmod q$ satisfies $p \ll q^L$ ($L \leq 5$, Heath-Brown 1992). This bounds primes in AP by the AP modulus, complementing the Burgess bound which bounds by the character modulus.

6.7 Salez (2014)

Salez verified the Erdős–Straus conjecture computationally up to 10^{17}. The present work provides a structural divisor-residue framework for explaining why small A values frequently succeed.

7 Summary of Results

Result	Status	Scope
$T \in D_A$ iff A works	proven	prime $n \equiv 1 \pmod 4$, $\gcd(n, A) = 1$
$-1 \in H(A) \Rightarrow -1 \in D_A$	partially proven + computational	prime A
Prime A: $T \in D_A$ iff a QNR prime factor exists	converse proven; forward conditional + computational	prime A
$n \equiv 5 \pmod 8$: $A = 3$ works	proven	prime $n \neq 3$
Composite n with factor $\equiv 3 \pmod 4$	claimed via Mballa (preprint)	such composites
$\left(\frac{p}{A}\right) = \left(\frac{n}{p}\right)$ for $p \mid m_A$	proven	applicable primes
$A \leq 31$ covers $n \leq 100{,}000$	verified	tested class
$A \leq 99$ covers all but one up to 10,000,000	verified	tested class
$A = O(n^{1/(4\sqrt{e})+\varepsilon})$	conditional proposition	analytic route
$A = O((\log n)^2)$	conditional on GRH	analytic route
$A \leq C$ for all n	conjecture	open

8 Open Problems

1. **Direct $T \in D_A$ criterion.** The original -1-route ($-1 \in H(A) \Rightarrow -1 \in D_A$) fails in 821 tested cases and should not be treated as a general lemma. The correct open problem is to prove a direct criterion for $T \in D_A$, likely through the centered divisor-residue set $C_A(N) = \{\prod p_i^{\delta_i} \pmod A) : -e_i \leq \delta_i \leq e_i\}$. The centered criterion $h/2 \in \Sigma_A(N)$ is perfectly equivalent to $T \in D_A$ across all 14,244 tested cases with zero mismatches.

2. **Centered divisor-residue set and D_A closure.** The clean formulation uses the *centered divisor-residue set*. Let $N = nx = nm$ with factorization $N = \prod p_i^{e_i}$. A divisor $P \mid N^2$ has $P = \prod p_i^{\alpha_i}$ with $0 \leq \alpha_i \leq 2e_i$. The target condition $P \equiv -N \pmod A$ is equivalent to $PN^{-1} \equiv -1 \pmod A$, where $PN^{-1} = \prod p_i^{\alpha_i - e_i}$. Define the centered set $C_A(N) = \{\prod p_i^{\delta_i} \pmod A) : -e_i \leq \delta_i \leq e_i\}$. Then A works $\iff -1 \in C_A(N)$. **Open problem:** prove that when some prime factor of N is a QNR modulo prime A, the centered set $C_A(N)$ contains

-1. This symmetric formulation directly captures the exponent budget and is cleaner than the shifted-set approach.

3. **Constant bound conjecture.** Prove $A \leq C$ for some absolute constant.

4. **Covering system proof.** Prove that $\{3, 7, 11, 15, 19, 23, 31\}$ covers all n, not just $n \leq$ 100,000.

5. **Proposition 8 verification.** The proof combines the Burgess bound [**?**], partial summation, and PNT in AP [10, Thm 5.7] to show the least prime QNR $\equiv 3 \pmod 4$ satisfies $A \ll n^{1/(4\sqrt{e})+\varepsilon}$. Each step is sourced to standard references. The combination requires verification of error term interactions by an analytic number theorist.

6. m**-route sufficiency.** Prove that when the direct route fails, the m-route always succeeds for some A in the covering set.

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Computational tools: Python 3.13, SymPy 1.14.0. All computations verified with exact rational arithmetic. AI assistance: Brodie Foxworth (OpenClaw agent) performed computational verification, developed the divisor-residue criterion, and prepared this manuscript under the direction of the author.